Zscaler Digital Delivery Consultant Exam Guide

ZPA Delivery Specialist Exam Guide

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About the Zscaler Digital Delivery Consultant Exam Guide

This evaluation will be facilitated by an instructor who is qualified and experienced in deploying ZIA,ZPA and ZDX. Similar to the practice session, you will be assigned a typical deployment scenario and an unconfigured lab environment. You will be given the time needed to implement a solution that meets the requirements set out in the scenario. Your solution will be evaluated against a checklist of requirements and a reference implementation. 75% pass marks required to be certified as a ZCDS capstone project.

Scenario

Consider the scenario where Safemarch, a large global organization is transforming their security and network operations. To transition to a Zero-Trust Network architecture, Safemarch has deployed Zscaler solutions.
You now need to apply what you have learned to do further preparation for the rollout of Zscaler services across Safemarch.

Requirements for items that need to be configured include:

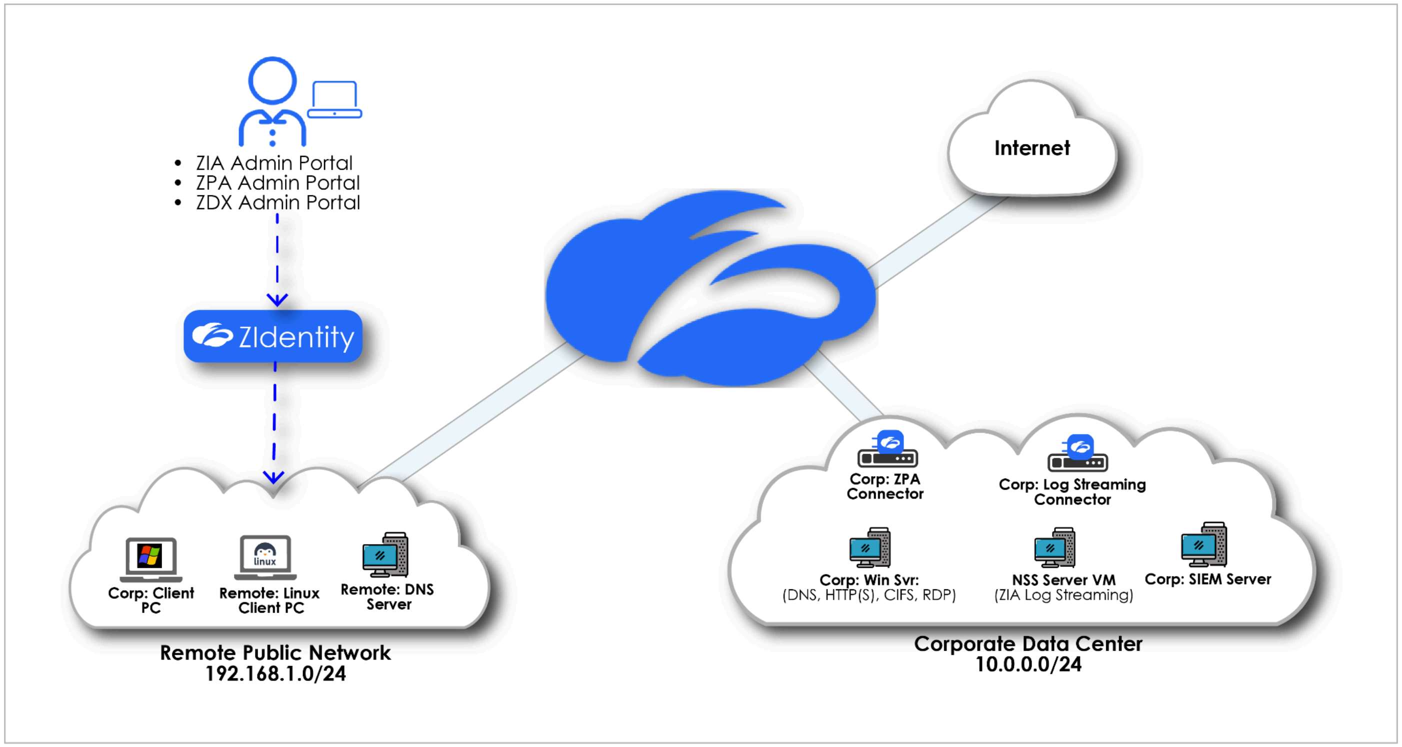
- ZIdentity admin entitlements for administrators to fully deploy and support users.
- ZIdentity service entitlements for end users to access needed services.
- Traffic Forwarding via Zscaler Client Connector and Zscaler App Connectors.
- Log Streaming & Reporting.
- Digital Experience Monitoring for users.
- Policies that enforce secure, policy-based access of employees to Safemarch internal applications.
- Privacy for Safemarch users accessing their private information while ensuring that all other traffic is inspected for threats and data leakage.

Lab Setup

For this lab each student is assigned a unique student user (e.g. student@zs**abcd**.safemarch.com)for access to a dedicated training pod of virtual machines (hosted in Skytap) with corresponding tenants.

Virtual Lab network topology

For your lab exercises, you will be configuring ZIdentity as your IdP to authenticate Zscaler Client Connector users, allowing them access to a range of public and internal corporate applications through the ZPA App Connectors on the Skytap corporate network.



Name	IP	Username	Password	Description
Corp: Client PC	192.168.1.1	Administrator	Admin-123!	Windows client
Remote: Linux Client PC	192.168.1.2	Admin	Admin-123!	Linux client
Remote: DNS Server	192.168.1.252	Administrator	Zscaler123!	DNS Server
Corp: Win Svr	10.0.0.8	Administrator	Admin-123!	Windows Server
Log streaming connector	10.0.0.20	admin	zscaler	App connector as ZPA log streaming
NSS Server	10.0.0.4	zsroot	zsroot	ZIA log streaming
Corp: SIEM Server	10.0.0.2	zscaler	Zscaler123!	Syslog server
ZPA Connector	10.0.0.3	admin	zscaler	Private app connectivity

Task 1: Configure Identity Services

In this task, you need to create organization specific departments and users and configure Administrator inactivity timeout duration.

Task Configuration Requirements

- Create departments, add users to respective departments.
- Use the Student admin account to configure Administrator Inactivity Timeout Duration to 180 minutes

User Name	Password	Department
marketing-<cd>@**zsabcd**.safemarch.com	Zscaler123!	Marketing
finance-<cd>@**zsabcd**.safemarch.com	Zscaler123!	Finance
itsecurity-<cd>@**zsabcd**.safemarch.com	Zscaler123!	IT Security
admin-<cd>@**zsabcd**.safemarch.com	Zscaler123!	Administrator
maintenance-<cd>@**zsabcd**.safemarcdch.com	Zscaler123!	Maintenance

Note:

1. Use the **Student Admin** for the **Assign Groups** and enable the **Skip Second Factor Authentication** option in the **Security Settings** to bypass MFA.

2. Ensure the Primary email address is set to be the user (**student@zsabcd.safemarch.com)** assigned to you in the Lab Access Information

3. Make sure to select the "**Set by Administrator**" option in the password option under **Security Settings**

4. Make sure to uncheck the option " Prompt **for Password Change After the Initial Log In**".

Where to get Help

If you need help with the details of each of the IdP configuration procedures, you can use the following links:

- https://help.zscaler.com/unified/adding-users-zidentity
- https://help.zscaler.com/zidentity/configuring-authentication-session

Task 2: Configure ZIA and ZPA Traffic Forwarding

In this lab, you need to prepare traffic forwarding configurations for Safemarch's Zscaler Client Connector users.

Configuration Requirements

Set up traffic forwarding to meet Safemarch's requirements, including:

1. Ensure when users are working from a remote location, all traffic from all ports and protocols, must be able to connect to Zero trust exchange.
2. Users must use a password to disable ZIA and ZPA.
3. Client Connector Log Mode must be "Error".
4. Client Connector must automatically install the Zscaler SSL certificate.
5. Client Connector configuration needs to be set to potentially allow traffic on any port and protocol to be forwarded.

Testing Requirements

- Login to Client PC VM, use student@zsabcd.safemarch.com account to login to Zscaler Client Connector and verify in Web Insights logs that the user is able to access the Internet.

Task 3 : Configure ZIA Policy

In this lab, you need to configure a set of policies that provide protection against security threats and control access to web-based applications, based on department membership.

Configuration Requirements

Configure recommended policy settings, implement a set of policies for the initial Safemarch deployment, including:

1. SSL inspect all traffic.
2. Exempt some specific web traffic from SSL inspection, including:
    a. Health and Government URL Categories to address privacy concerns
    b. A custom list of URLs that may be maintained by adding / removing URLs as needed to respond to SSL inspection-related issues.
    c. Ensure all other policies are enforced for traffic that is not SSL inspected.
3. For all departments, except IT Security, block Executable file downloads from all URL category.
4. Caution users in the IT Security department before the download of Executable files.
5. Finance users should be blocked for uploading all files except Microsoft office files.
6. All supported file types must be sandbox inspected. While files are being inspected, users are allowed to download them, except for files in the following URL categories that must be quarantined and inspected before allowing the user to download them if they are determined to be safe:
    a. Illegal or Questionable
    b. Security
    c. Miscellaneous
7. All traffic must be firewall inspected. Add specific rules to allow specific permitted access for:
    a. An approved list of network services which includes SSH, Kerberos and pc Anywhere users in the IT Security department
8. All users need to be granted access to all URL Categories, except for a default rule to block Legal Liability URL Categories that includes:
    i. Adult Material, Drugs, Gambling, Illegal or Questionable, Militancy/Hate and Extremism
    ii. A custom URL category for any additional URLs that may be deemed off limits by Safemarch security and access policies. (**Hint**: Add any arbitrary URL's as part of this new custom URL Category)

b. The customer's IT Security team needs to be able to get access to www.eicar.org site file downloads for continuous evaluation of security tools' effectiveness. Also block www.eicar.org for the rest of the users.
c. Block all users from using Social Networking and Online Shopping, EXCEPT employees in the Marketing department must be allowed to view and post to LinkedIn, Twitter and Facebook
d. Office 365: Forward all O365 traffic and enable Microsoft Recommended One-Click configuration
e. Enable advanced policy settings for URL & Cloud App Controls, including:
    i. Suspicious New Domains Lookup
    ii. AI/ML based Content Categorization
    iii. Embedded Sites Categorization
    iv. Enforce SafeSearch

9. Allow login to only the Microsoft O365 Safemarch corporate tenant and prevent personal login to Microsoft O365 products like OneDrive and Outlook.

    Tenant Profile Name: **Safemarch Microsoft O365**
    Tenant Directory: **deadbeef-0001-ffee-0001-314159265358**
    Office 365 Tenant: **safemarch.com**

10. Block access to all streaming media except the Safemarch YouTube channel.
    The Network Security Team at Safemarch has provided the following firewall rules from their legacy appliance solution for reference:

- Rule Name: *Allow Safemarch YouTube Channel*
  Criteria:
    a. Policy: Allow
    b. URL Category: Safemarch YouTube Custom Category
        i. Sites: ytimg.com, .ytimg.com
        ii. Regular Expressions:
            1. www\.youtube\.com/.UCSylwuqCXM_W13ARfzASm3Q
            2. youtube\.com/videoplayback

3. www\.youtube\.com/crossdomain\.xml
4. youtube\.com/generate

Ensure the functionality of these legacy rules is properly 'translated' into ZIA policies to block access to all streaming media except the Safemarch YouTube channel.

- To retrieve the Youtube Channel ID, check the Configuration Hint section.

Testing Requirements

- Check Web Insights logs and certificate info on various URLs to confirm SSL inspection/exemptions
- Verify access to some legal liability URL, e.g. www.freeslots-machines.com is blocked
- Verify access to online shopping sites is blocked for all users
- Verify access to social networking sites is blocked for users **NOT** in the Marketing department
- Check Firewall logs and make sure SSH traffic is hitting the intended firewall rule
- Go to youtube.com and try to watch some video to confirm that it is blocked
- Go to some streaming media site, e.g. vimeo.com, or netflix.com and verify these sites are blocked

Configuration Hints

YouTube Channel ID:

YouTube channel ID is a unique identifier (e.g., UCxxxxxxxxxxxxxxxxxxxxxx), distinct from channel name/handle. It is often used for API analytics, 3rd party integrations etc., It's a unique 24-character string. You may use any of the following methods to retrieve the Channel ID:

1. Method 1: Using the Page Source:
   a. Open the YouTube channel (https://www.youtube.com/@ZscalerInc) in Chrome browser.
   b. View Page Source: Right-click on the page, select View Page Source. You can also press Ctrl+U on Windows / Opt+Cmd+U on Mac to open the same.
   c. Search with following strings:
      i. "og:url" content=
      ii. https://www.youtube.com/channel/
      iii. The string starting with "UC" (24-characters long) is the Channel ID.

2. Method 2: Using 3rd party tools:

   The following utilities can also retrieve the Channel ID

   a. https://commentpicker.com/youtube-channel-id.php
   b. https://seostudio.tools/youtube-channel-id
   c. https://www.tunepocket.com/youtube-channel-id-finder/

Note: The 3rd party utilities worked at the time of creating this document. There is no guarantee that they will always work in the future.

Task 4: Configure ZIA Logging & Reporting

Configure log streaming to meet Safemarch's logging requirements to send logging information to an onsite SIEM for long-term retention and integration into the overall security information infrastructure. For this deployment you will initially configure ZIA to stream logs to a SIEM server to test the configuration.

> **Note:** In this scenario, a NSS Server VM (10.0.0.4) and a SIEM Server VM (10.0.0.2) have already been pre-installed in your Skytap environment.

Configuration Requirements

In this lab, you need to configure Nanolog Streaming Service (NSS) to accommodate Safemarch's need for long-term storage of and real-time streaming of web logs to a SIEM. You will configure to stream the following log data:

a. Web logs (Tab-separated format)
b. Time zone: America/New York
c. Duplicate Logs will be sent for 60 minutes
d. Include all records for web transactions for users in the Finance and Marketing departments.

Testing Requirements

- On the **Corp: Client PC**, generate traffic that matches the NSS configuration.
- Log in to the Corp: SIEM Svr and verify that log messages are being received in /var/log/syslog.
    - *Note*: You can use PUTTY on the Corp: Win Svr to connect to the Corp: SIEM Svr.

Configuration Hints

- Start the syslog, check the status.
    - sudo systemctl start rsyslog
    - sudo systemctl status rsyslog
- View the syslog entries as they are received.
- When configuring the NSS Server’s **Management Interface** IP address, use: **10.0.0.4/24**
- When configuring the NSS Server’s **Service Interface** IP address, use: **10.0.0.5/24**
- For the **Nameserver** IP address, use **10.0.0.254**
- For the **Default Gateway** IP address, use **10.0.0.254**

- TCP Port use **514**

- To transfer the NSS Certificate to the NSS Server VM, you can use **WinSCP** which is pre-installed on the **Corp: WinSvr**.

Task 5: Configure ZPA Policy

Configuration Requirements

Implement a set of policies for the initial Safemarch deployment, including:

1. Provision App connector Corp: ZPA Connector
2. Allow all employees access to the following private applications:
    a. Intranet: **intranet.patraining.safemarch.com**, Use TCP port 80, 443.
    b. File Share: **shares.patraining.safemarch.com**. Use TCP port 139, 445.
3. Allow access to a server via **SSH** only on domain-joined devices. Use TCP port 22.
4. Only users in the **Maintenance** and **Administrator** departments should be able to access **hvac.patraining.safemarch.com**. Use TCP port 80.
5. All users excluding users from the **Administrator** Department are to be blocked from using Safemarch's remote desktop application: **rdp.patraining.safemarch.com**. Use TCP and UDP port 3389.
    a. Blocked users should get a notification.
    b. Ensure that idle, open connections are closed after 15 min.
    c. RDP connections require re-authenticate every 10 min.
6. Limit access to devices on 10.0.100.0/24 across any port to the **Administrator** Department.
7. Auto-discover any other applications that are running on the Safemarch corporate network.

Testing Requirements

- To transfer the provisioning key to the App Connector, you can use WinSCP on the Corp:Win Svr VM.
- For the purposes of this lab only, it is sufficient to enable one App Connector in the data center.

Note: If the provisioning of an App Connector fails, re-provision it as follows:

1. Stop the App Connector service using the command **sudo systemctl stop zpa-connector**
2. Delete all files in the relevant folder with the command **sudo find "/opt/zscaler/var" -mindepth 1 -delete**
3. Reboot the VM using the command reboot
4. Stop the App Connector service again (it auto-starts on boot) using the command **sudo systemctl stop zpa-connector**
5. Copy back the Provisioning Key file using the command **sudo cp provision_key /opt/zscaler/var/**
6. Start the App Connector service again using the command **sudo systemctl start zpa-connector**
7. Check App Connector status using the command **sudo systemctl status zpa-connector**

- Check the status of the App Connector.
- To test access to the corporate file share application, access **\\shares.patraining.safemarch.com** from the Client PC
- To test RDP access, you can simply open the application from the Windows taskbar and enter **rdp.patraining.safemarch.com**
- Check **Diagnostics** logs and **Live** Logs in the ZPA Admin Portal to verify that your policies have the desired effect.

Task 6: Configure ZPA Logging & Reporting

In this lab, you need to configure Log Streaming Service (LSS) to accommodate Safemarch's need for long-term storage of and real-time streaming of web logs to a SIEM.

- Configure the **Corp: Log Streaming Connector** and to stream logs to a simulated SIEM server.
- A SIEM Server VM has already been pre-installed in your environment.

Configuration Requirements

configure log streaming to meet Safemarch's logging requirements to send logging information to an onsite SIEM for long-term retention and integration into the overall security information infrastructure.

1. Configure the App Connector Corp: Log Streaming Connector. This is used for streaming logs from ZPA cloud to SIEM server.
2. You will configure LSS to stream the following log data:
    a. User Activity logs for Client Connector users only (CSV format)
    b. App Connector Status logs (CSV format)

Testing Requirements

On the Corp: Client PC, generate traffic that matches the LSS configuration.

- Log in to the SIEM VM and verify that log messages are being received.
    - *Note*: You can use PUTTY on the Client PC VM to connect to the SIEM Server VM.
- Ensure SIEM server can be accessed via ZPA.

Configuration Hints

- To upload the provision key to the App Connector, you can use **WinSCP** which is pre-installed on the **Corp: Win Svr VM**.
- The TCP Port for the Log Receiver is **514**.

Task 7: Prevent the leakage of PII and PCI data

Configuration Requirements

Implement a set of configurations to meet the lab task requirements, including:

1. Ensure that all file upload traffic is fully inspected to ensure that the confidential PCI data is blocked and the activity logged. Attempt to upload the files to dlptoolbox website should fail.
2. After the file upload is blocked by DLP policy, extract the relevant event logs from Web Insight. The logs are exported in csv format. The logs should only contain the entries related to the DLP block event. The following columns should be selected: Event time, User, Policy Action, URL, DLP MD5, DLP Identifier, Blocked Policy Type, DLP Dictionary, DLP Engine.
3. Save the Exported file to the desktop in Client PC VM.

Testing Requirements

1. Login with the student@zsxxxx.safemarch.com credentials to Zscaler Client Connector on the Corp: Client PC and verify that Internet Security status is enabled in the Zscaler Client Connector.
2. Open https://dlptoolbox.com on a browser and confirm access to the website.
3. Attempt to upload the **"Customer Credit Card.xlsx"** and "**Confidential SSNs and Credit Cards.xlsx"** files from the **DLP Test Files** folder on the desktop in the File Audit pane on the dlptoolbox website.
4. Confirm that a block notification is displayed when attempting the upload.
5. Check Web Insights logs in the ZIA Admin Portal to ensure the file upload attempt is logged and blocked.

Where to Get Help

Refer to the following resources for guidance on configuring and testing DLP policies:

- Web Insights Logs: Filters

Configuration Hints

1. Be sure to use predefined dictionaries i.e. **Credit Cards** while creating the DLP engine.

Task 8: Apply Zero Trust Automation Requirements : Identify Blocked Traffic

Monitor web traffic logs to identify potentially malicious requests, such as access to restricted URLs or connections from suspicious IP addresses and alert the administrators while blocking the source.

Configuration Requirements

1. Fetch and analyze the web URLs and identify the allow/deny list by using the advanced endpoint (**{{ZIABase}}/security/advanced**)of the **ZIA Base URL**. For e.g. Deny or backlist a list of sites, programs, or other elements that are not allowed to be started or visited for your organization. Allow up to 25000 URLs. (**Sample URLs**: tinder.com", "snapchat.com", "gamixlabs.com", "888.com")
2. Block unauthorized URLs (betano.com", "bet365.com)using the **POST /security/advanced/blacklistUrls** endpoint **({{ZIABase}}/security/advanced/blacklistUrls?action=ADD_TO_LIST)**
3. Notify administrators of the incident using **POST /dlpNotificationTemplates** (**input:** Name, Subject and the Plain Text Message)
4. Validate the updates

Testing Requirements

- Create an API client in ZIdentity Dashboard
- Authenticate a session for the OneAPI client in Postman
- Call the API endpoint to add the malicious URLs
- Navigate to the ZIA Admin portal and see if the URLs are added to the blocked list
- Notify the Admin about the update

Where to get Help

If you need help with the details of each of the policy configuration procedures, you can use the following links:

- https://help.zscaler.com/zia/security-policy-settings#/security/advanced/blacklistUrls-post
- https://help.zscaler.com/zia/data-loss-prevention#/dlpNotificationTemplates-post

Task 9: Adding Custom Application and Creating Custom Probes

Add a custom application and create probes (**Web and Cloud Path**) that log data pertaining to certain aspects of the applications in your organization.

Configuration Requirements

- Add a custom application in the ZDX Admin portal
- Add a web and cloud path probe for the custom application you created. Use values of your choice for probing criteria and exclusion criteria.